

WAVESHARE ESP32-S3 — DIRECT 120VAC WIRING TO DOMETIC BRISK II

SSR-25DA for Compressor | Relay Channels for Fans + Furnace | Dual Freeze Protection

WAVESHARE ESP32-S3-RELAY-6CH

IP65 Enclosure | 7-36V DC
Relay: 10A @ 250VAC per channel

CH1 (GPIO 1): Furnace

CH2 (GPIO 2): Compressor→SSR

CH3 (GPIO 41): Fan Low

CH4 (GPIO 42): Fan High

CH5 (GPIO 45): Available

CH6 (GPIO 46): Available

I2C: GPIO 8/9

DS18B20: GPIO 10

GND

12V+

3.3V

FURNACE

(Dry Contact Closure)

Separate unit with own blower

T1

T2

SSR-25DA

Solid State Relay
3-32VDC → 25A/380VAC

DC+

DC-

CH1 NO

AC1 (in)

AC2 (out)

120VAC HOT

HOT

NEUTRAL

LOAD CHAIN

Supco SFPC (freeze stat)
NC: opens 35F, closes 50F

Bimetal Cutout

Fan Hi (CH4 → Pin 2)

Fan Lo (CH3 → Pin 4)

120VAC SOURCE

6-PIN DOMETIC CABLE

to Brisk II Rooftop AC
(14 AWG minimum)

Pin 1: Blue (Compressor)

Pin 2: Black (Fan Hi)

Pin 3: Yellow (unused)

Pin 4: Red (Fan Lo)

Pin 5: White (Neutral)

Pin 6: Green (Ground)

DS18B20 FREEZE SENSOR

Waterproof probe on evaporator coil

Software: Cut compressor @ 32F, restart @ 45F

4.7KΩ pull-up: Data → 3.3V

SAFETY CRITICAL — 120VAC WIRING

CH2 relay NO switches 12VDC milliamp trigger to SSR DC+
SSR-25DA switches 120VAC to compressor (heatsink required ~15W @ 12A)
CH3/CH4 switch 120VAC directly to fans (2-3A, 10A relay rating)
CH1 = dry contact for furnace (separate unit)
Dual freeze protection: DS18B20 software (32F) + Supco SFPC hardware (35F)
All 120VAC wiring: 14 AWG minimum, proper junction box per NEC

COMPONENT SPECIFICATIONS

WAVESHARE ESP32-S3-RELAY-6CH:

Power: 7-36V DC (or 5V USB-C) | IP65 (6.3" x 4.33" x 3.54")
Relay: 10A @ 250VAC/30VDC per channel (6 channels)

GPIO ASSIGNMENTS:

CH1=GPIO1 CH2=GPIO2 CH3=GPIO41 CH4=GPIO42 CH5=GPIO45 CH6=GPIO46
I2C: SDA=GPIO8, SCL=GPIO9 | 1-Wire: GPIO10 | Buzzer: GPIO21 | LED: GPIO38